

HarvestLink Dispatch Desk

Musanze Cooperative Harvest and Dispatch Decision Lab. A proposed staff dashboard. This document does not implement the screen. It shows how a dispatch officer reads the pipeline and decides.

Group code AI-G03. The same code is printed in the dashboard header and on every result.

1. Cover, people, and the operational problem

Role	Name	Registration
Member 1 — Data and UX lead	Nuzha ZainEl-Abdeen Mohammed Ismail	25/27419
Member 2 — Regression engineer	Ehab Fakhralden Mohamed Hamid	25/27950
Member 3 — Classification engineer	Mojtaba Abdalitieef Ahmed	25/27660
Member 4 — Clustering and QA engineer. Group leader.	Mohammed Osama Hasan	25/27014
Member 5 — Reproducibility and release lead	Lazarus Simboya Ira Inyasio	25/28180
Prediction and demonstration lead	Feras Saifaddin Ismail Mohammed	25/27916

Stakeholder. The dispatch officer at Musanze HarvestLink Cooperative. The officer is not a modeller. The work happens at a collection desk, sometimes with strong window light, on a laptop. The next action has to be readable without interpreting a chart legend first.

Problem. Before a potato consignment leaves, the officer needs three answers from the issued batch, then a human decision. The model proposes. The officer dispatches, holds, or overrides.

- 1. **Expected harvest weight.** An estimate in kilograms from the six farm measures. It is not a scale reading.
- 2. **Dispatch attention.** Whether this consignment should be held for a look before it leaves.
- 3. **Operating profile.** Which other collection rows look similar on those same measures. A profile is not a verified farm type.

What Member 1 owns. The schema check, the feature matrix, data_report.json, and this dashboard. The matrix uses plot area, rainfall, soil pH, seed, distance, and arrival hour. record_id stays a label. actual_yield_kg and dispatch_attention stay targets and are not inputs. Pages 2 to 7 are the owned design. Later members fill the numbers the screens reserve.

Where each required part sits

Page	What the assessor can check
2	The path from CSV upload to review, prediction, and the officer's decision
3	Data-quality gate, batch list, and expected weight
4	Dispatch warning and the profile plot
5	Uncertainty, missing data, model limitation, and human override
6	Accessible colours, type, labels, and legends
7	Three justified design decisions

From the CSV to a human decision

Primary user. Dispatch officer on the HarvestLink desk.

Goal. Confirm the file, read the three results for a consignment, and record Dispatch, Hold, or Override.

Constraint. Figures come from the pipeline for group AI-G03. The officer cannot rename columns or type over a result. A failed file check shows no weight, warning, or profile.

1. Upload
Officer chooses the issued CSV. The header already shows AI-G03.

2. File check
Fingerprint, row count, missing cells, duplicate ids.

3. Results
Expected kg, attention word, and profile for each complete row.

4. One row
The officer opens a consignment and reads all three together.

5. Decide
Dispatch, Hold, or Override with the model still visible.

Happy path

1. Open the desk. Read AI-G03 in the header.
2. Upload the group CSV. The screen shows the SHA-256 of those exact bytes and the file name.
3. Read the file check. Six features are named in words. Missing cells and duplicate ids are counts, not colours alone.
4. Open the batch. Each complete row has an expected weight, an attention word, and a profile word.
5. Open one consignment. The six measures are listed beside the three results and the model version.
6. Record the decision. Dispatch and Hold need one click. Override needs a written reason. The result stays on the record.

Paths the officer will actually hit

- **Wrong table.** A missing or extra column stops the desk. The message names the columns. No results are drawn.
- **Empty cell.** That row is listed as incomplete and has no weight, flag, or profile. Complete rows remain reviewable.
- **Repeated record id or repeated row.** The count is shown on the file check. Both copies stay visible so the officer can compare them.
- **Bad attention label.** A value other than 0 or 1 stops the file check and names the field.
- **Single-record form.** A missing or non-numeric measure blocks prediction and names the field. The other fields stay filled.
- **Close call.** When the attention score sits between 40 and 60 in 100, the chip reads Uncertain. Hold remains the filled button.

What moves between steps

Step	Officer sees	Comes from
File check	Group code, fingerprint, row count, feature count, missing values, duplicate counts, complete and incomplete counts	data_report.json
Batch and detail	Expected kilograms, attention word plus score, profile word, model version	Regression, classification, and clustering outputs
One new record	The six measures in, and the same three results out. A bad field returns a validation message and no JSON result.	predict.py
Decision	Dispatch, Hold, or Override, plus reason when overriding	The officer. Never the model alone.

What stays off the decision

Field	How the desk treats it
record_id	Shown as Identifier. It is never a measure in the estimate, the warning, or the profile.
actual_yield_kg	Used to teach the weight model. The desk shows the estimate, not a box for editing the past weight.
dispatch_attention	Past labels stay inside the file. The officer acts on the new warning and does not retype 0 or 1.
Fingerprint	Shown in full on the file check so this batch can be matched to the issued file.

File check and expected weight

Figures marked Sample are layout only. The running pipeline replaces them. The fingerprint is calculated from the uploaded file and is never a fixed string in the design.

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AI-G03

File check · AI_A1_G03.csv

Rows

Sample

Features

6

Incomplete

Sample

Duplicate ids

Sample

Fingerprint

64 hex characters from this file

Missing cells

Count beside each column name

Duplicate rows

Count, with both copies kept

Column check

Passed

Review results

Consignment

AI-G03 · model version on the record

Identifier

Sample-014

Plot area

1.2 ha

Rainfall

81 mm

Soil pH

5.7

Seed

210 kg

Distance

14 km

Arrival hour

09:00

SAMPLE

Expected harvest weight

— kg

Estimate from the six measures above. Not a scale reading.

File check. This is the gate. Review results stays unavailable until the column names and the attention labels pass.

The six features are written as plot area, rainfall, soil pH, seed, distance, and arrival hour. The record id is labelled Identifier, not a measure.

Column statistics stay behind “Show statistics”. The officer’s first task is to trust the file, not to read a quartile table.

A failed check replaces this body with the column message on page 5. The button is not merely greyed out.

Expected weight. The number is the regression prediction for this row. The unit kg is written beside it.

The six measures above the weight are the inputs. Past measured yield is not shown as something to edit. It was used to teach the estimate.

Under the number, a second line can show the typical miss from the held-out check, in kilograms, once that metric exists. The line is absent until the file is real.

The example measures match the single-record shape in the brief: 1.2 ha, 81 mm, pH 5.7, 210 kg seed, 14 km, hour 9.

Batch the officer scans before opening one row. Incomplete rows stay in the list and cannot be chosen for dispatch.

Identifier	Expected kg	Attention	Profile	Next step
Sample-014	Filled by the model	Hold	Profile 2	Open
Sample-015	Filled by the model	Ready	Profile 1	Open
Sample-016	No result	Incomplete	No profile	Blocked

Dispatch warning and profiles

Dispatch attention

AI-G03

Hold

SAMPLE SCORE

This consignment needs a look before it leaves. The score is a sample layout, written both as words and as a count out of 100.

Sending a load that needed a check is harder to undo than holding a load that was ready. Hold stays available either way.

Hold

Dispatch anyway

Override

Attention warning. The chip always contains the word Hold, Ready, or Uncertain. Colour repeats the word. It does not replace it.

The score from the classifier is shown as “n in 100” and as a sentence, for example “Hold is more likely than Ready.”

On a Hold or Uncertain row the filled button is Hold. Dispatch anyway is an outline. Override asks for a reason on the next step and does not clear the warning.

Operating profiles

AI-G03 · k chosen from 2 to 5

Profile groups similar measurements. A profile is not a verified farm type.

○ Profile 1

□ Profile 2

△ Profile 3

This row	Profile 2 · square on the plot
Why this k	Silhouette comparison for k = 2, 3, 4 and 5. The winning k is named in a sentence.
Plot	Every complete record is one mark. The legend stays beside the plot.

Profiles. Labels are Profile 1, Profile 2, and so on, up to the selected k. This drawing shows three marks only as a layout sample.

Each profile has a shape and a colour. The shape remains if the colour washes out. The table repeats the profile in words.

The screen does not rename a profile “good farms” or “problem farms”. A short measure summary may say which inputs sit higher than the batch, and that sentence stays descriptive.

Incomplete rows appear in a list under the plot and have no profile mark.

Uncertainty, gaps, limits, and override

These four states are part of the desk. A result screen without them is incomplete.

Uncertainty
Uncertain
When the attention score is from 40 to 60 in 100, the chip reads Uncertain even if the model leans one way. The sentence says the two outcomes are too close. Hold is the filled button. The score remains visible as a count out of 100.

Missing data
Stopped
File: "Rainfall is missing on 2 rows. Those rows have no weight, warning, or profile." Single record: "Rainfall must be a number. Nothing was predicted." The named field is the only one cleared of its error when the officer fixes it.

Model limitation
A standing note sits under every result. "This estimate uses six measures from the group file. It does not know the road, the sacks, or today's queue. A profile is not a farm type." The note stays when the numbers look confident.

Human override
Override opens a reason box: "Why are you overriding the suggestion?" Save stays unavailable until the reason has a short sentence. The stored record keeps the model weight, the attention word, the profile, the model version, AI-G03, the officer's decision, and the reason.

Stopped file

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AI-G03

This file does not match the harvest table. Missing columns and unexpected columns are listed by name. No weight, warning, or profile was calculated.

Choose another file

Override, with the suggestion kept

Your decision
AI-G03 · suggestion still Hold

Model suggestion **Hold** stays on this record.

Reason	Why are you overriding the suggestion?
Decision	Dispatch · attributed to the officer, not the model

Save decision **Back**

Back returns to the consignment without saving. The batch list is unchanged.

Single record the officer types

Check one consignment
AI-G03

Rainfall	Rainfall must be a number. Nothing was predicted.
Other five measures	Stay as the officer typed them.

Predict **Clear**

Colour, type, and legends

The desk uses dark ink on warm paper so the text still reads in window light. Colour supports a word. It never replaces the word.

Primary #243B53 Header and filled button. White label.	Paper #F6F3EC Screen background.	Surface #FFFFFF Cards and fields.	Ink #1A1A1A Body text.
Muted #5C574E Labels and footnotes.	Ready #1B6B3A Always with the word Ready.	Uncertain #8A5A00 Always with the word Uncertain.	Hold / stop #9B2335 Always with Hold or the error words.

Type

The desk uses the system font so browser zoom can enlarge it. Text wraps. Chips grow with their words instead of clipping.

Role	Size	Weight	Use
Screen title	22	700	Desk name, consignment title
Result figure	28	700	Expected kilograms only
Body	16	400	Sentences, field values, reasons
Meta	14	400	Group code, fingerprint, model version

Legend and status rules

- Attention chips: Ready, Hold, Uncertain. Each has a word, a border, and a fill.
- Profile marks: circle, square, triangle, diamond, and a cross if k is 5. The legend lists Profile plus the number. The plot and the table use the same pair.
- One filled button per decision row. Hold is filled when the chip is Hold or Uncertain. On a Ready row, Dispatch can be the filled button, and Hold stays one click away.
- Buttons are at least 48 by 48. Gap between two buttons is 8. Screen padding is 16. Cards use 8 px corners.
- Error text uses Hold red and the sentence. A red border without words is not a message.

Contrast pairs the desk relies on

Ink on surface	Where
#1A1A1A on #FFFFFF and on #F6F3EC	Body text and field values
#FFFFFF on #243B53	Header title, group code, and the filled button
#9B2335 on #FFFFFF, with the words of the message	Stop, Hold, and field errors
#5C574E on #FFFFFF	Footnotes, fingerprint label, model version

Why the desk is arranged this way

Decision 1 — The file check is a gate

The officer sees the fingerprint, the column check, the missing cells, and the duplicate counts before any weight. Review results does not open on a failed table. Justified because a later file can keep the column meanings and change the rows, and a familiar file name is not proof. Acting on a weight from the wrong table is the failure this screen is built to prevent.

Decision 2 — Hold is the filled action on a warning

When the chip says Hold or Uncertain, the filled button is Hold. Dispatch anyway is secondary. Override keeps the suggestion on the record and requires a reason. Justified because a load that leaves and later proves it needed a check is harder to recover than a load kept at the desk. The officer can still send it. The model does not send it.

Decision 3 — Profiles stay numbered

Clusters are Profile 1 upward, with a shape and a colour, plus the sentence that a profile is not a verified farm type. The desk does not invent names such as “weak farms”. Justified because the same grouping can mix different causes, k is only the silhouette choice from 2 to 5, and a category name would overclaim what the plot can show.

Parts the desk must keep

Part	Looks like	Rule
Header	Slate bar, white title, AI-G03 under the title	On every screen, including errors and override
File check	Four counts, fingerprint row, Passed or Stopped	Results stay hidden until Passed
Weight block	Kilograms, then the estimate sentence	Never labelled as a scale reading
Attention chip	Ready, Hold, or Uncertain	Word plus colour plus score out of 100
Profile legend	Shape, colour, Profile number	Same mapping on the plot and the row
Decision row	One filled button and two outline buttons	Override requires a reason

Accessibility checks

- Body text stays dark on white and on warm paper. The filled button is white on slate.
- Every status the officer acts on includes the word Ready, Hold, Uncertain, Passed, Stopped, or Incomplete.
- The profile plot can still be read from the shape and the number when colour fades.
- Text uses the system font and wraps. Zoom must not clip a chip or a button label.
- The main actions are at least 48 by 48, including Hold, Dispatch anyway, Override, and Save decision.

Member 1 evidence. Nuzha ZainEl-Abdeen Mohammed Ismail owns this design and the data functions in `src/data_pipeline.py`. Export this file as `AI_A1_G03_UIUX.pdf`. The group code on every screen is AI-G03.